Detecting short gravitational-wave bursts in real LIGO noise

Frequentist, Bayesian and machine-learning analysis of simulated bursts injected into GWOSC O1 data

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Abstract

We use eight stretches of public LIGO-Hanford strain from the first observing run (O1) as a source of *real*, non-Gaussian, non-stationary noise, inject three families of unmodelled bursts (sine-Gaussian, Gaussian pulse, linear chirp) at a known optimal SNR, and compare three ways of detecting them at the same false-alarm probability: a frequentist matched-filter bank (profile-likelihood ratio), a Bayesian analysis of the same likelihood (exact grid posterior, MCMC, Bayes factors) and machine-learning detectors (feature-based classifiers, a convolutional network and its ensemble, an autoencoder). All methods share the data, the injections and a split by chunk of data. (i) The whitened noise is Gaussian in the bulk but has a glitch floor of a few 10^{-3} per second and large epoch-to-epoch variations; they dominate the cost of low false-alarm probabilities and make the calibration of a threshold epoch-dependent. (ii) At a false-alarm probability of 10^{-2} per one-second window the matched-filter bank detects sine-Gaussians with 50 % efficiency at an optimal SNR of ≈ 5 (the oracle that knows the parameters: 3.0), and the Bayes factor detects the same bursts as the bank, with calibrated posteriors and an explicit prior dependence. (iii) A small CNN ensemble reaches parity with the bank on sine-Gaussians and is about one SNR unit better on the pulses and chirps it was trained on, but loses efficiency on a morphology it has not seen; its probabilities are well calibrated and its uncertainty flags the windows not to be trusted.

1 Introduction

Since the first detection of a binary black-hole merger [1] gravitational-wave detectors have produced a stream of strain data that is public through the Gravitational Wave Open Science Center (GWOSC) [2]. Searches for *short, unmodelled transients* (bursts) cannot rely on a bank of astrophysically motivated waveforms and must either use phenomenological templates or make no assumption on the signal shape (excess-power and coherent-network methods [3, 4]). They all face the same statistical problem: the noise is only approximately Gaussian and stationary, the signals are weak ($\rho \sim 5$–10), the parameters of the signal are unknown nuisance parameters, and the required false-alarm probability is tiny.

This project uses that problem as a laboratory for the techniques of the course *Statistics and Data Analysis*: hypothesis testing and the Neyman–Pearson lemma, maximum likelihood, profile likelihood and Wilks' theorem, toy Monte Carlo, Bayesian inference with priors, MCMC and Bayes factors, supervised and unsupervised learning, calibration and uncertainty in machine learning. The organisation follows the earlier projects on LHCb open data (asymmetry in $B^{\pm} \to K^{\pm}K^{+}K^{-}$) and on atomic-force-microscopy curves: one shared data set, one notebook

per method, one common set of figures of merit, a report that says where each number comes from.

Research question. *At a fixed false-alarm probability on real LIGO noise, how do a frequentist matched-filter bank, a Bayesian analysis of the same likelihood and machine-learning detectors compare in detection efficiency, in the quality of their uncertainty statements, and in robustness to the non-Gaussian noise and to signal morphologies they were not designed for?*

Scope. Single detector (H1); three phenomenological burst morphologies injected at a chosen optimal SNR; false-alarm probabilities per one-second window. This is *not* an astrophysical sensitivity estimate: there is no second detector, no coincidence and no time-slide background, and SNRs are not converted into distances.

Organisation. Section 2 describes the data, the whitening and the tests of the noise model. Section 3 defines the signals, the SNR and the injection chain. Sections 4, 5 and 6 are the three analyses; Section 7 compares them and Section 8 gives conclusions and limits. Appendix A maps every number to the notebook and result file that produced it.

2 Data and noise model

2.1 Selection of the data

The data are LIGO-Hanford (H1) strain at 4096 Hz from O1 (September 2015–January 2016), downloaded with `GWpy` [5] from GWOSC. Eight chunks of 4096 s were selected, evenly spread over the run, fully inside the H1 science segments and at least 2000 s from every catalogued event (GW150914, GW151012, GW151226 and the two candidates); the closest chunk is more than 1.7×10^5 s from an event, so the chunks can be treated as noise only. Table 1 lists them with the train/validation/test assignment. The split is made *by chunk*, interleaved in time so that each part samples the whole run, never by window: neighbouring windows share noise. The test chunks (3 and 7, Nov. 2015 and Jan. 2016) therefore come from detector epochs, PSDs and glitch conditions that no threshold or model has seen.

chunk	start (UTC)	split	windows
0	2015-09-12 06:59	train	4056
1	2015-10-04 16:01	val	4056
2	2015-10-19 13:17	train	4056
3	2015-11-07 23:46	**test**	4056
4	2015-11-28 01:03	train	4056
5	2015-12-16 19:37	val	4056
6	2015-12-31 06:38	train	4056
7	2016-01-19 13:43	**test**	4056

Table 1: The eight H1 chunks (4096 s each). Windows are the non-overlapping one-second analysis windows left after removing 20 s at each end.

2.2 PSD, whitening and analysis windows

The noise power spectral density $S_n(f)$ of each chunk is estimated with Welch's method [6] (4 s Hann segments, 0.25 Hz resolution) using the *median* average, which is robust to glitches. Each

chunk is whitened with its own PSD, band-passed to 30–500 Hz and down-sampled to 1024 Hz (the band is below the new Nyquist frequency, so no information is lost), then cut into non-overlapping windows of one second (1024 samples). One window is the unit of every later analysis. In the frequency domain a window has bins $V_k = \mathrm{rfft}(v)_k/\sqrt{512}$ at 1 Hz spacing; for stationary Gaussian noise $\mathrm{Re}\, V_k$ and $\mathrm{Im}\, V_k$ are independent $\mathcal{N}(0,1)$ in the band, so that the inner product of two signals is $\langle a, b\rangle = \mathrm{Re}\sum_k A_k B_k^*$ and the optimal SNR is

$$\rho^2 = \langle h, h\rangle = 4\int \frac{|\tilde{h}(f)|^2}{S_n(f)}\, df. \tag{1}$$

Figure 1 shows the amplitude spectral density of the eight chunks: the shape is stable, the in-band sensitivity of chunks 1–7 is within 7 % of chunk 0, and the usual lines (60 Hz and harmonics) are visible.

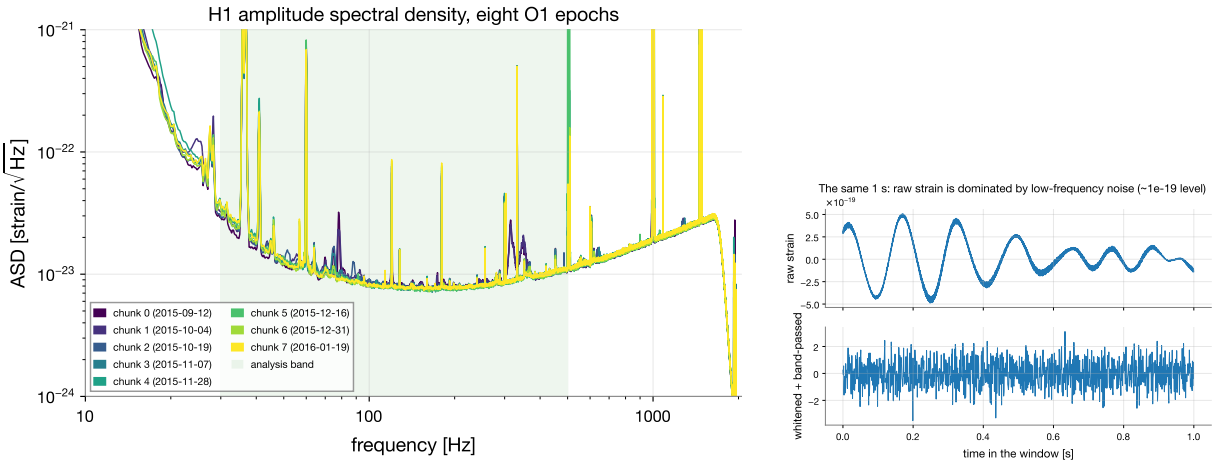


Figure 1: Left: ASD of the eight chunks. Right: the same second of data, raw and whitened (the raw strain is dominated by low-frequency noise).

2.3 How Gaussian and stationary is the noise?

The likelihood used in Sections 4 and 5 assumes independent standard-normal whitened bins. We test it against a Gaussian reference built with the same chain from ideal bins (Figure 2):

- **Bulk.** The sample distribution follows $\mathcal{N}(0,1)$ down to a probability of $\sim 10^{-4}$; the bin variance is 2 (the expected value) within a few per cent in the quiet epochs, and the correlation between adjacent bins is 0.012, at the level of the statistical floor.
- **Lines and non-stationarity.** The bin variance dips at the 60 Hz harmonics (the median-Welch PSD slightly over-estimates narrow lines at 1 Hz resolution) and shows broad excess power in the glitchier epochs (chunks 4 and 6).
- **Tails.** A fraction 10^{-4} of the samples lies beyond 5σ (Gaussian: 6×10^{-7}) and the loudest windows reach hundreds of σ; the plain standard deviation of chunk 6 is 1.8 against a robust σ of 0.95. Chunk 2 is exceptional: 3.5 % of its windows exceed the Gaussian 99.9th percentile of the window energy and the lag-1 autocorrelation of the window energy is 0.80 (long glitchy stretches).

These are the glitches and the non-stationarity of the real detector. They are the reason why *every* threshold in this report is set on real background, and why calibration is always tested on chunks that were not used to set it.

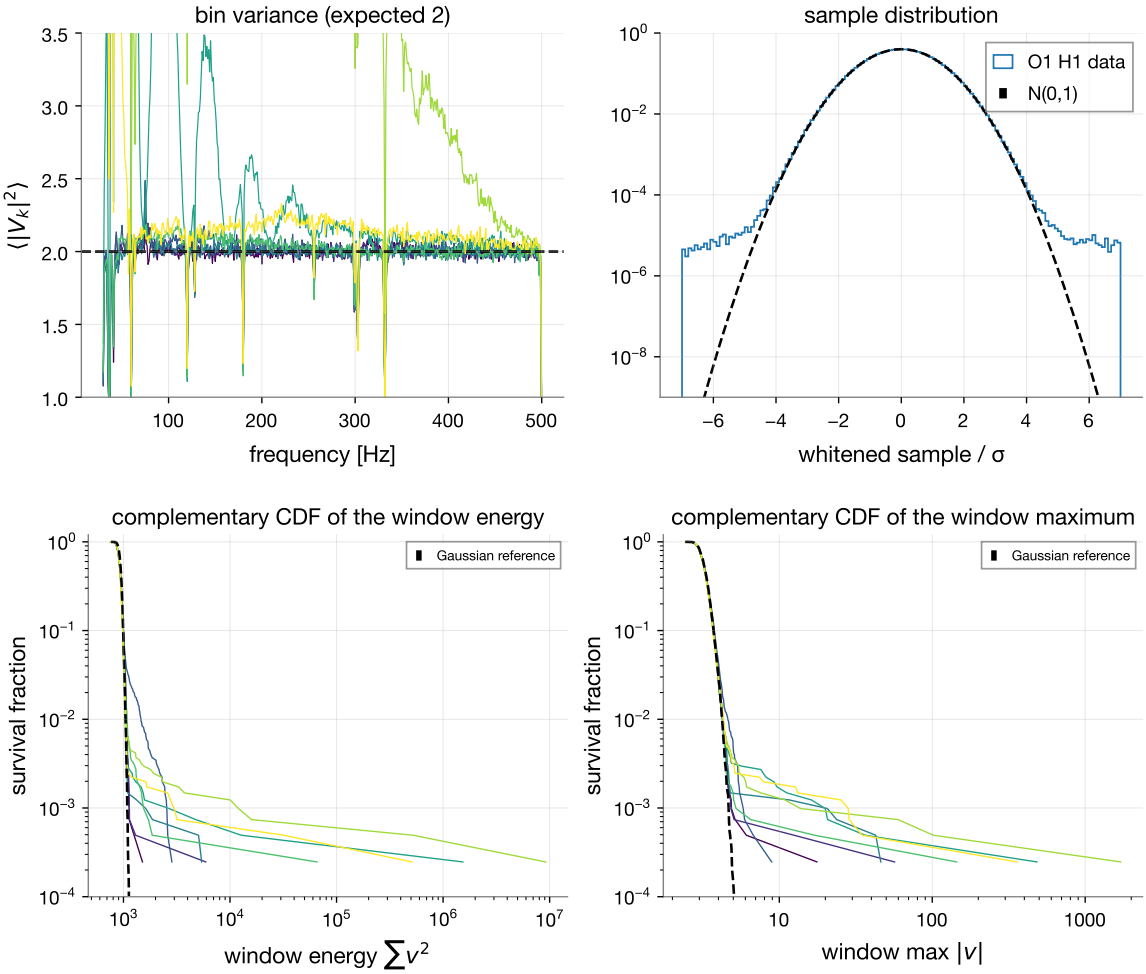


Figure 2: Top: bin variance of each chunk (left; the expected value is 2) and distribution of the whitened samples against $\mathcal{N}(0,1)$ (right). Bottom: complementary distribution of the window energy and of the window maximum for each chunk against the Gaussian reference (dashed).

3 Burst signals, SNR and injections

3.1 Morphologies

Three phenomenological morphologies are injected (Table 2); all have closed-form Fourier transforms (Gaussian integrals with complex width), which are implemented exactly and checked against a brute-force numerical transform (agreement to 10^{-4} or better for all three).

3.2 Injection chain and validation

Signals are not added in the time domain after whitening. They are built directly as whitened frequency bins $V_k = 2\tilde{h}(f_k)/\sqrt{S_n(f_k)}$ with the PSD of the chunk of the host window and added to the real whitened bins. This is equivalent to injecting in the strain (whitening is linear), gives the exact optimal SNR of Eq. (1) and has no filter transients. The exploratory notebook used the time-domain proxy "L_2 norm of the whitened waveform over the noise standard deviation"; it agrees with Eq. (1) up to the constant factor 1/0.95 (the noise standard deviation) for all morphologies, and was replaced by the analytic definition.

The chain is validated on real noise: injecting sine-Gaussians at $\rho = 0, 5, 10$ in 4000 test windows

morphology	$h(t)$	injection prior
sine-Gaussian (SG)	$e^{-t^2/2\sigma^2}\cos(2\pi f_0 t+\phi)$, $\sigma = Q/2\pi f_0$	$f_0 \sim$ log-U(50,400) Hz, $Q \sim$ U(4,15)
Gaussian pulse	$e^{-t^2/2\sigma^2}$	$\sigma \sim$ log-U(0.8,5) ms
linear chirp	$e^{-t^2/2\sigma^2}\sin(2\pi f_0 t+\pi \dot{f} t^2+\phi)$	$f_0 \sim$ U(50,150) Hz, $\dot{f} \sim$ U(150,600) Hz/s, $\sigma \sim$ U(30,80) ms

Table 2: The three burst morphologies. For all of them $t_0 \sim \mathrm{U}(0.25,0.75)$ s inside the one-second window and $\phi \sim \mathrm{U}(0,2\pi)$.

and matching with the exact template, $|z|^2$ follows a non-central $\chi^2_2(\rho^2)$ law (means 2.03, 27.1, 101.8 against 2, 27, 102; Kolmogorov–Smirnov $p = 0.64, 0.66, 0.49$). Figure 3 shows what the injections look like.

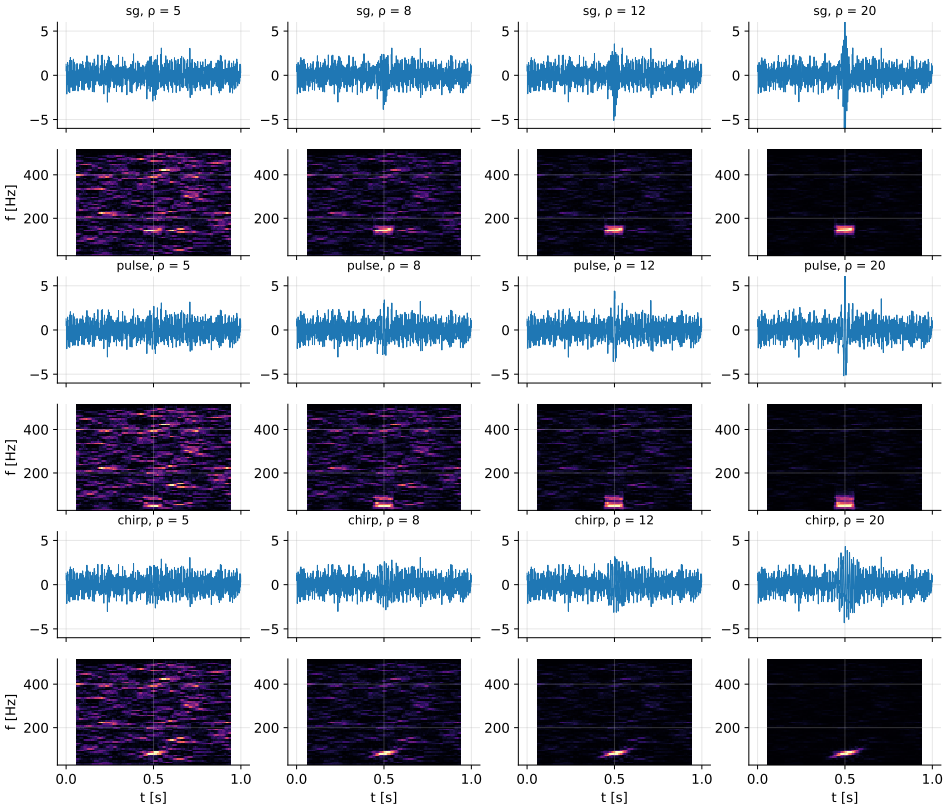


Figure 3: The three morphologies injected into the same real noise window at $\rho = 5, 8, 12, 20$: whitened series (top row of each pair) and spectrogram (bottom). At $\rho = 5$ nothing is visible by eye.

4 Frequentist analysis

4.1 Hypotheses and test statistic

We test H_0 (noise only) against H_1 (noise plus a sine-Gaussian burst). For a window with whitened bins X and a unit-norm complex template $T(t_0, f_0, Q)$ the signal is aT with $a = a_c + ia_s$ complex (amplitude and phase), and $\ln \mathcal{L} = -\frac{1}{2}\|X - aT\|^2$ up to constants. Maximising

analytically over a gives the likelihood-ratio statistic

$$\lambda(t_0, f_0, Q) = -2 \ln \frac{\mathcal{L}(0)}{\mathcal{L}(\hat{a})} = \rho^2(t_0, f_0, Q) = \Big| \sum_k X_k T_k^* e^{2\pi i k t_0} \Big|^2, \qquad \lambda_{\max} = \max_{t_0, f_0, Q} \rho^2, \tag{2}$$

evaluated for a bank of 160 templates (40 log-spaced $f_0 \in [40, 450]$ Hz $\times$ $Q \in \{3, 6, 10, 16\}$) and for all $t_0 \in [0.15, 0.85]$ s (the time shift by one FFT). At a *fixed* template Wilks' theorem [7] applies: $\lambda \sim \chi^2_2$ under H_0 (two degrees of freedom: the two quadratures). For $\lambda_{\max}$ it does not: (t_0, f_0, Q) exist only under H_1, which is Davies' problem [8] and the look-elsewhere effect. Two other statistics are used for comparison: the *oracle*, $|z|^2$ at the true parameters (the Neyman–Pearson optimal test of a simple hypothesis, not available in practice), and an *excess-power* statistic [3], the maximum over time and over five durations (10–200 ms) of the z-scored band energy, which assumes nothing about the morphology.

4.2 Background: Wilks, trials factor and real noise

Scanning 20 000 windows of ideal Gaussian noise, $\lambda_{\max}$ is accurately described by $N_{\text{eff}} = 6.8 \times 10^3$ independent χ^2_2 trials, $P(\lambda_{\max} > x) = 1 - (1 - e^{-x/2})^{N_{\text{eff}}}$: a single-template threshold of $\lambda \approx 9$ at FAP 10^{-2} becomes ≈ 27 for the bank. On real noise (Fig. 4) the distribution follows the Gaussian one up to $\lambda_{\max} \approx 22$ and then leaves it, developing a nearly flat *floor* at a few 10^{-3} per window: the glitches of Section 2. The thresholds set on the training chunks are 33.6 (FAP 10^{-2}) and 114 (FAP 10^{-3}), against ≈ 27 and ≈ 31 for Gaussian noise with the same trials factor. Epochs differ enormously: $\sim 10\,\%$ of the windows of chunk 2 exceed $\lambda_{\max} = 60$, and this chunk drives the training threshold.

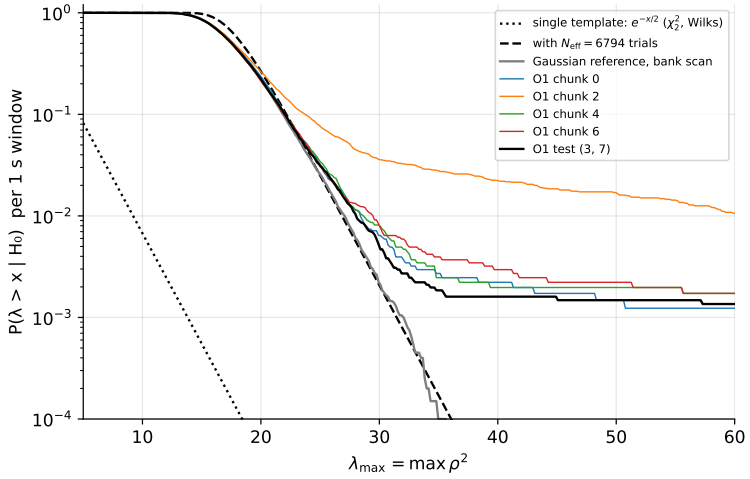


Figure 4: Survival function of $\lambda_{\max}$ under H_0: single-template χ^2_2 (dotted), χ^2_2 with $N_{\text{eff}} = 6794$ trials (dashed), Gaussian simulation (grey), real chunks (colours) and test chunks (black).

4.3 Calibration on unseen epochs

The threshold at false-alarm probability α is the $(1 - \alpha)$ quantile of the training background. On the test chunks the observed FAP of the bank is 2.2×10^{-3} for $\alpha = 10^{-2}$ (68 % interval 1.7–2.9×10^{-3}) and 1.2×10^{-3} for $\alpha = 10^{-3}$; the excess-power statistic gives 3.7×10^{-3} and 1.1×10^{-3}. The thresholds are therefore *conservative*, by a factor ≈ 4.5 at 10^{-2}, because the glitchy chunk 2 inflates the training background. The false-alarm probability is not a property of the statistic but of the (non-stationary) background it is calibrated on; a threshold transferred to another epoch can be off by a factor of several in either direction.

4.4 Efficiency and ROC

For each morphology and SNR, 500 bursts are injected in randomly chosen test windows (with the PSD of their own chunk) and the fraction above threshold is the efficiency (Fig. 5, Table 3).

- **Sine-Gaussian.** The oracle reaches 50 % efficiency at $\rho = 3.0$ and 90 % at 4.4; the bank at 5.5 and 6.9; excess power at 6.8 and 9.7. The oracle–bank gap is the price of not knowing (t_0, f_0, Q) (look-elsewhere); the bank–excess-power gap is the price of not using the signal shape.
- **Pulse and chirp.** Outside the template family the bank is mismatched but still better than the agnostic statistic: the Gaussian pulse is well covered by low-frequency, low-Q templates ($\rho_{50} = 5.9$ against 6.1), and the chirp loses only 1.7 in ρ_{50} with respect to the sine-Gaussian (7.2) while excess power needs 10.8.
- **FAP 10^{-3}.** Because of the glitch floor, lowering the FAP tenfold costs much more than in Gaussian noise: the 50 % point of the bank moves from $\rho \approx 5.5$ to ≈ 11 (dotted curves).

The ROC AUC at $\rho = 8$ is 0.997/0.971 (bank/excess power) for sine-Gaussians, 0.995/0.994 for pulses and 0.983/0.820 for chirps.

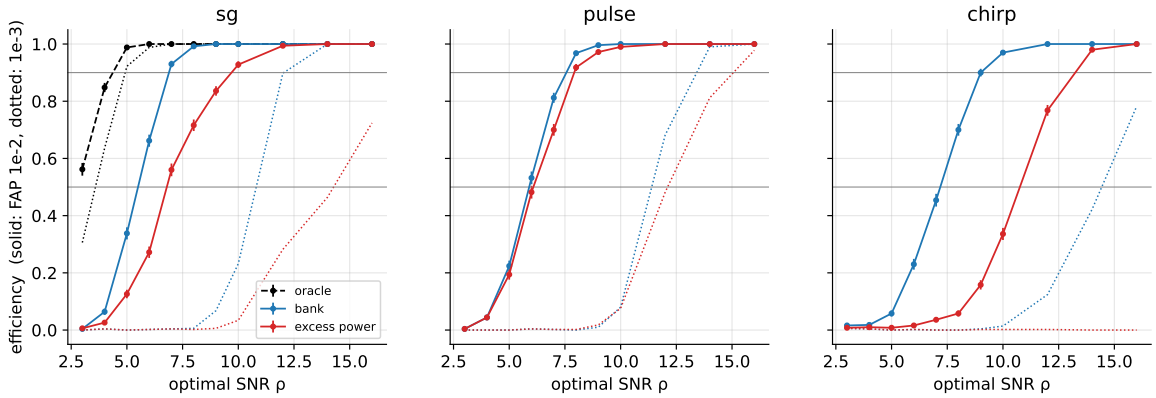


Figure 5: Efficiency against optimal SNR at FAP 10^{-2} (solid, with binomial errors) and 10^{-3} (dotted) for the three morphologies: oracle (black, sine-Gaussian only), bank (blue), excess power (red).

morphology	statistic	ρ_{50}	$\pm$	ρ_{90}	$\pm$
sine-Gaussian	oracle	3.00	0.00	4.37	0.09
	bank	5.50	0.07	6.89	0.05
	excess power	6.79	0.08	9.70	0.14
Gaussian pulse	bank	5.90	0.07	7.56	0.08
	excess power	6.08	0.10	7.92	0.06
linear chirp	bank	7.19	0.09	9.00	0.12
	excess power	10.76	0.09	13.25	0.10

Table 3: Optimal SNR at which the efficiency reaches 50 % and 90 % (FAP 10^{-2}, test chunks).

4.5 Maximum-likelihood parameter estimation

For detected sine-Gaussians the bank maximum is refined with `iminuit` on the continuous profile $\rho^2(t_0, f_0, Q)$ and the covariance of the MLE is compared with the Fisher (Cramér–Rao) matrix, obtained from the Hessian of $-\rho^2$ at the truth on noise-free data, $\mathrm{cov} = 2[\nabla^2(-\rho^2)]^{-1}$. With 300 injections per SNR (Fig. 6) the MLE saturates the bound for $\rho \gtrsim 10$ (f_0: RMS 2.8 against 2.7 Hz at $\rho = 10$; 0.90 against 0.90 Hz at $\rho = 30$) and the Hessian errors of `iminuit` agree with it. At low SNR the RMS departs upward (at $\rho = 6$ the RMS of f_0 is 34 Hz against 4.5 predicted)

because the likelihood surface has secondary maxima to which the estimate jumps; the robust width stays near the bound. This is the classical threshold effect of non-linear parameters.

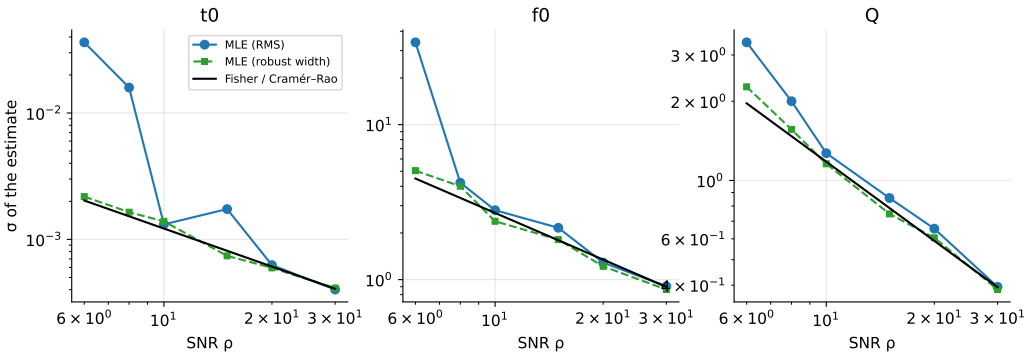


Figure 6: Standard deviation of the MLE of t_0, f_0, Q against SNR: RMS (blue), robust width (green) and Fisher/Cramér–Rao bound (black).

4.6 Toy Monte Carlo: bias, pulls and coverage

With a known template and unknown SNR $s \geq 0$ the data reduce to $z = s\,e^{i\varphi} + n$, $n \sim \mathcal{N}(0, I_2)$; the MLE is $\hat{s} = |z|$ (Rice distribution) and $-2\Delta \ln \mathcal{L}_p(s) = (\hat{s} - s)^2$, so the profile-likelihood interval is $[\hat{s} - 1, \hat{s} + 1]$ truncated at zero. With 4×10^4 toys per value of s (Fig. 7): the MLE is biased upward at low SNR ($E[\hat{s}] - s = 1.25$ at $s = 0$, $\sqrt{\pi/2}$; 0.27 at $s = 2$; 0.05 at $s = 10$); the profile-likelihood interval covers 39 % at $s = 0$ (nominal 68 %), overshoots to $\approx 74\,\%$ near $s = 1$ and is within one per cent of 68 % only for $s \gtrsim 3$ (parameter on the boundary: Wilks' regularity fails); the Neyman central and Feldman–Cousins belts [9] cover 68.3 % at every s by construction. The same experiment repeated with a fixed template injected in 5000 real windows per SNR gives the same bias, RMS and coverage within 0.03, so the single-template amplitude estimate is insensitive to the glitches (whereas the bank statistic of the previous section is not).

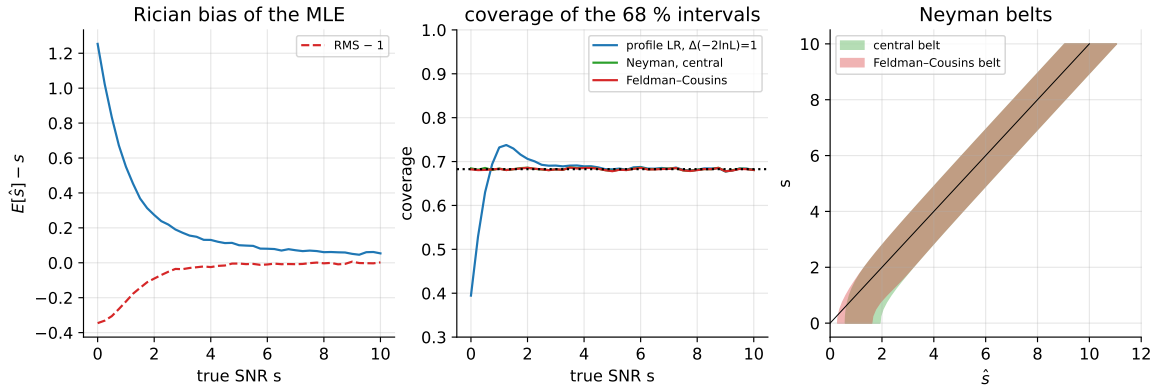


Figure 7: Toy Monte Carlo of the amplitude estimate. Left: bias and RMS − 1 of $\hat{s}$. Centre: coverage of the 68 % profile-likelihood, central Neyman and Feldman–Cousins intervals. Right: the two Neyman belts.

4.7 Systematic study

The observable is the sine-Gaussian efficiency $\varepsilon(\rho = 8)$ at FAP 10^{-2}; each variant changes one analysis choice and recomputes the whole chain on the *same* windows and injections, so that differences are correlated. Barlow's criterion [10] (a variation is significant if $|\Delta|/\sigma_\Delta > 2$, $\sigma_\Delta^2 = |\sigma_{\mathrm{var}}^2 - \sigma_{\mathrm{nom}}^2|$) gives Table 4. Only the use of the *mean* instead of the median in the Welch average

is large and unambiguous (+0.025): the mean PSD is inflated by glitches, lowers the whitened-noise level and the threshold (42.9 against 45.0). All other variants change ε by at most 0.007 (1–3σ). We quote 0.025 as the systematic uncertainty on $\varepsilon(\rho = 8)$.

variant	threshold	ε	Δ	$\lvert\Delta\rvert/\sigma_\Delta$ (Barlow)
nominal	45.0	0.928	–	–
PSD mean-average	42.9	0.953	+0.025	5.3
PSD 2 s segments	44.6	0.935	+0.007	2.8
PSD 8 s segments	44.6	0.933	+0.005	2.4
band 20–500 Hz	45.0	0.929	+0.001	1.1
band 30–400 Hz	45.0	0.925	−0.003	1.9
coarse bank (20 f_0)	44.0	0.921	−0.007	2.9

Table 4: Systematic variants (threshold from chunks 0 and 2, efficiency from 1000 injections in chunks 3 and 7).

5 Bayesian analysis

5.1 Model and exact marginal likelihood

The Bayesian analysis uses the same Gaussian likelihood as Section 4 with priors on the signal parameters (Table 5). The two linear amplitudes $a = a_c + ia_s$ get independent Gaussian priors, so that the SNR $\rho = |a|$ is Rayleigh(σ_ρ); the default is $\sigma_\rho = 10$. With $z(\theta) = \sum_k X_k T_k^* e^{2\pi i k t_0}$ and $\rho^2(\theta) = |z|^2$ (the matched-filter output of the frequentist analysis at fixed $\theta = (t_0, f_0, Q)$) the integral over a is a Gaussian integral:

$$\ln \frac{p(X|\theta,\mathcal{M}_1)}{p(X|\mathcal{M}_0)} = \frac{1}{2}\frac{\sigma_\rho^2}{1+\sigma_\rho^2}\,\rho^2(\theta) - \ln(1+\sigma_\rho^2), \qquad a\,|\,\theta, X \sim \mathcal{N}\Big(\tfrac{\sigma_\rho^2}{1+\sigma_\rho^2} z,\ \tfrac{\sigma_\rho^2}{1+\sigma_\rho^2}\Big). \tag{3}$$

The closed form agrees with a brute-force two-dimensional integration over a to 10^{-13} ($\ln B = 10.7017$ both ways). The posterior of θ and the Bayes factor B_{10} then need only the matched-filter cube $\rho^2(t_0, f_0, Q)$, i.e. exactly the object computed in Section 4: the two analyses differ only in how the cube is combined (maximum against prior-weighted average) and in the prior.

parameter	prior
t_0	uniform on $[0.25, 0.75]$ s (as the injections)
f_0	log-uniform on $[50, 400]$ Hz
Q	uniform on $[4, 15]$
a_c, a_s	$\mathcal{N}(0, \sigma_\rho^2)$ each, $\sigma_\rho = 10$

Table 5: Priors of the Bayesian analysis.

5.2 Exact grid posterior and MCMC cross-check

The posterior is evaluated exactly on a grid of 96 log-spaced $f_0 \times 12$ values of $Q \times 512$ time steps (5.9×10^5 cells, uniform prior per cell by construction); the SNR posterior is the mixture over cells of Rice densities. Figure 8 shows one noise window with the same sine-Gaussian injected at $\rho = 5, 8, 12, 20$: at $\rho = 5$ the (f_0, Q) posterior is broad, from $\rho = 8$ the time of arrival is determined to a millisecond and at $\rho \geq 12$ everything collapses around the truth; the noise-only window gives a nearly flat t_0 posterior and an SNR posterior piled up at small ρ (the prior speaking). An

independent `emcee` [11] run (five parameters, no marginalisation, 40 walkers × 6000 steps) agrees with the grid on means and widths for $\rho = 8, 12, 20$ (e.g. $f_0 = 151.9 \pm 2.4$ against 152.0 ± 2.4 Hz at $\rho = 8$); the integrated autocorrelation time is 260–380 steps (acceptance 0.32–0.38), i.e. 420–610 effective samples. One difference is informative: at $\rho = 20$ the grid width of f_0 is 1.65 Hz against 1.14 Hz from MCMC, because the posterior is narrower than the grid cell (2.3 %, i.e. 3.4 Hz); at high SNR the grid is the limiting factor.

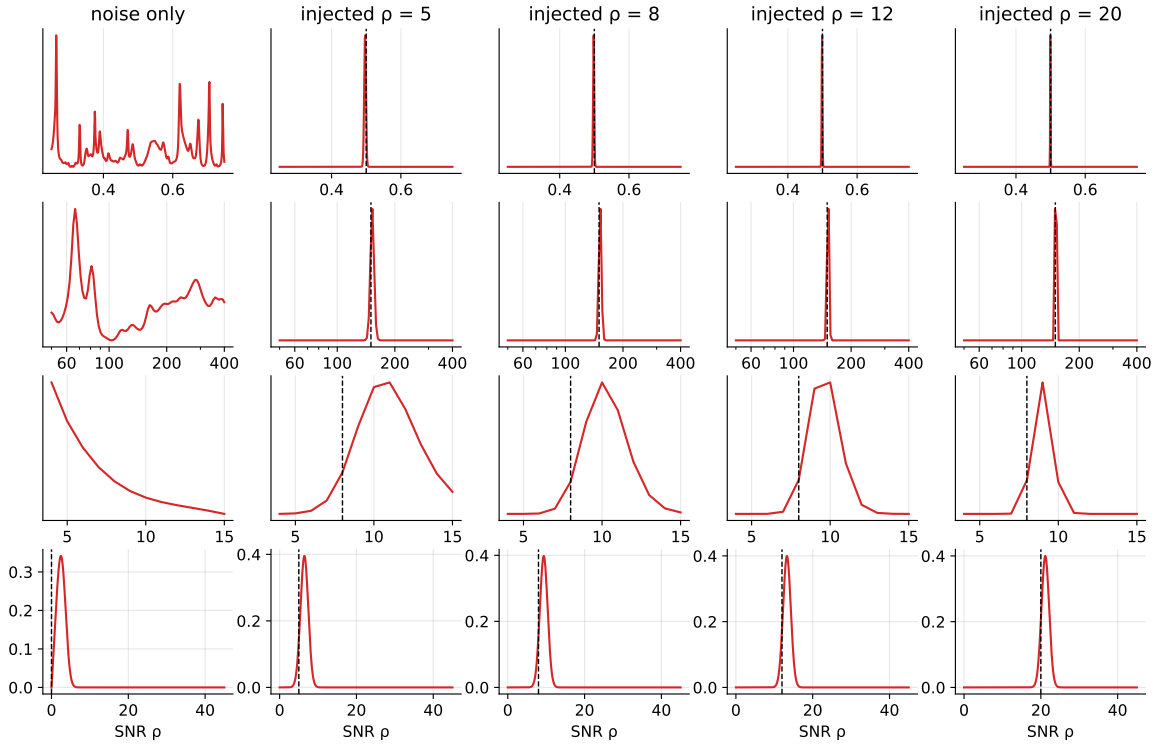


Figure 8: Posterior marginals of t_0, f_0, Q and ρ for the same noise window with no signal and with a sine-Gaussian ($t_0 = 0.5$ s, $f_0 = 150$ Hz, $Q = 8$) injected at $\rho = 5, 8, 12, 20$; dashed: truth.

5.3 Bayes factor as a detection statistic

We compute $\ln B_{10}$ with the same 160-template bank as the frequentist statistic, on the same background and injections. Three facts:

- **On real noise the Bayes factor is not a false-alarm probability.** A fraction 1.2 % (0.3 %) of the training (test) noise windows has $B_{10} > 10$ and 0.9 % (0.17 %) has $B_{10} > 100$: glitches are evidence for *something* that the Gaussian model can only call a burst. The Jeffreys scale [12] is a scale of evidence within the model; the false-alarm probability has to be calibrated on background exactly as for the frequentist statistic.
- **Same detection power.** At equal FAP (10^{-2}) the efficiencies of $\lambda_{\max}$ and $\ln B_{10}$ coincide within 0.01 at every SNR ($\varepsilon(\rho = 6) = 0.718$ against 0.714, $\varepsilon(\rho = 8) = 0.994$ for both) and so do the ROC AUCs (0.9763 against 0.9769 at $\rho = 6$; 0.9968 for both at $\rho = 8$): the ranking of the windows is almost identical.
- **The prior matters for B_{10}, not for the ranking.** At fixed data the Bayes factor falls as $-2 \ln \sigma_\rho$ for large prior width (Occam penalty for a prior that wastes mass on SNRs the data exclude): for a noise window it decreases monotonically from 0 to −7 between $\sigma_\rho \to 0$ and 100, for a signal it has a maximum at σ_ρ of the order of the true SNR (≈ 5 for $\rho = 8$, ≈ 10 for $\rho = 12$); the range of the t_0 prior shifts it by the logarithm of the volume ratio (Fig. 9). Whenever a Bayes factor is quoted the prior must be quoted with it.

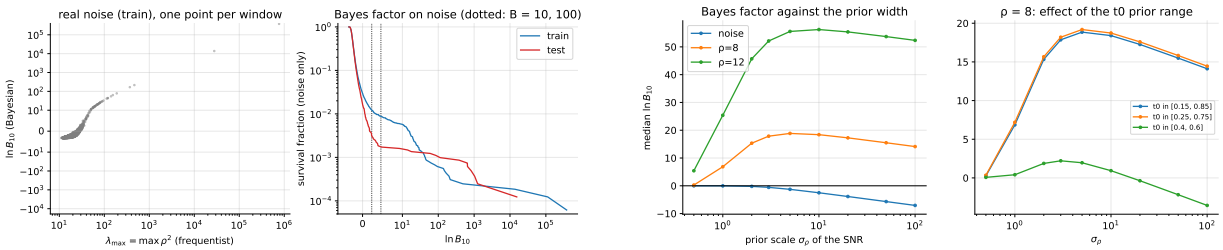


Figure 9: Left: $\ln B_{10}$ against $\lambda_{\max}$ on real noise (one point per window) and its tail. Right: median $\ln B_{10}$ against the prior width σ_ρ for noise and sine-Gaussians at $\rho = 8, 12$; and for $\rho = 8$ against the range of the t_0 prior.

5.4 Calibration of the posterior

Simulation-based calibration [13] draws the truth from the prior, simulates data, and checks that the posterior CDF of the truth is uniform. With 500 simulations in ideal Gaussian noise and 500 in real test windows, the rank histograms of t_0, f_0, Q and ρ are consistent with uniformity in both (Kolmogorov–Smirnov $p = 0.24$–0.90 in Gaussian noise, 0.25–0.85 in real noise), so that waveforms, whitening convention, closed-form marginalisation and grid are correct end to end. Five hundred simulations are too few to see an effect of the glitches ($\lesssim 1\,\%$ of windows). The coverage of the nominal 68 % credible intervals is 0.65–0.69 for Q and ρ but 0.50–0.51 for f_0 and 0.58–0.60 for t_0, in both noises: this is a grid-resolution effect (for loud injections the posterior is narrower than a cell: 3.4 Hz in f_0, 1 ms in t_0), not a model failure, and the MCMC does not have it. A posterior-predictive check on the residual of the fit (400 injections at $\rho = 10$) gives 0.25 % of p-values below 0.01 and 2 % below 0.05 (median 0.69 instead of 0.5, because the in-band noise variance is 1.975 rather than 2 per component).

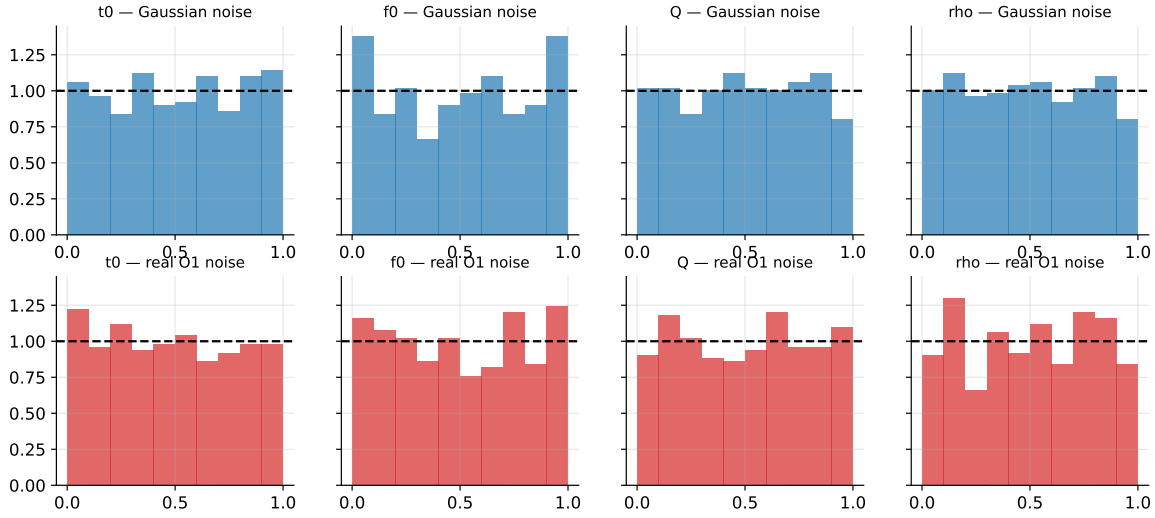


Figure 10: Simulation-based calibration: distribution of the posterior CDF of the true parameters (top: Gaussian noise; bottom: real O1 noise). Uniform = calibrated.

5.5 Credible against confidence intervals at low SNR

In the simplest setting where the two can be compared exactly (known template, $s \geq 0$ measured through $\hat{s} = |z|$) we compute the frequentist coverage of the 68 % credible intervals by integrating the Rice density (Fig. 11). For $s \gtrsim 3$ credible and confidence intervals coincide at 68–69 % (the posterior becomes Gaussian and the prior irrelevant). Below that, credible intervals — which

are conditional on the observed data and make no coverage promise — behave differently: with a flat prior the central interval covers 0 at $s = 0$, the HPD interval 77 % at $s = 0$ and 85 % near $s = 1$ before dipping to 53 % at $s \approx 1.7$; with the Rayleigh(10) prior both under-cover for $s \lesssim 1.5$ (54–60 % at $s = 1$) because the prior prefers larger SNRs than the truth. The Neyman/Feldman–Cousins intervals cover by construction, and the profile-likelihood interval lies in between (39 % at $s = 0$, 74 % at $s = 1$). The HPD intervals are shorter than the profile ones for $\hat{s} \lesssim 2$: informativeness is traded against guaranteed coverage.

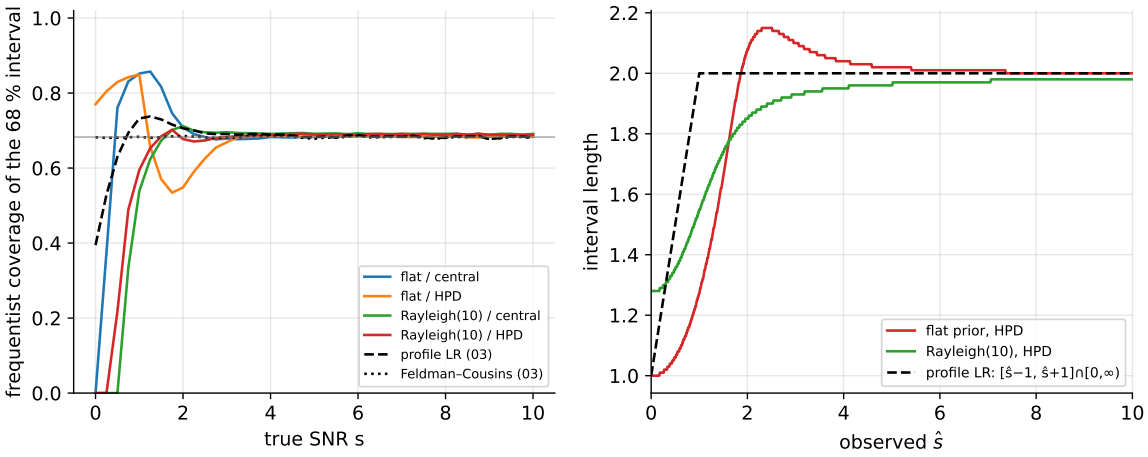


Figure 11: Left: frequentist coverage of the nominal 68 % intervals against the true SNR (credible intervals with two priors and two orderings; profile likelihood and Feldman–Cousins from Section 4). Right: interval length against the observed $\hat{s}$.

6 Machine-learning detectors

6.1 Data set and models

The labelled set is built from the same chunks and the same split: for each noise window of a split we also build a signal window, the *same* window with a burst whose morphology is drawn uniformly among the three and whose optimal SNR is uniform in $[3, 15]$. Classes are balanced and the label-0 windows contain all the real glitches and non-stationarity (16 224 training and 8112 validation windows per class, 8112 test windows). A second training set contains only sine-Gaussians, to test generalisation to morphologies not seen in training. The networks see the whitened series clipped to ± 12 (glitches reach hundreds of σ); the feature pipeline uses the unclipped series.

Feature-based models. Twenty-two morphology-agnostic features: the maximum over time of the z-scored sliding energy at five durations (the excess-power statistic at each scale), the maximum and the standard deviation over time of the energy in seven frequency sub-bands (30–500 Hz, from a short-time spectrogram), and $\max |v|$, kurtosis and std/MAD of the window. No matched-filter output is used.

CNN. A one-dimensional convolutional network of 17 761 parameters: three Conv–BatchNorm–ReLU blocks (kernels 15, 9, 5; 16, 32, 64 channels; max-pooling by 4), global max-pooling (translation invariance: the burst can be anywhere), a dense layer with dropout 0.2 and one logit; binary cross-entropy, Adam [14], batch 128, early stopping on the validation loss (14 epochs).

The *ensemble* is five independently initialised copies (the logits are averaged); a second network is trained on sine-Gaussians only.

Autoencoder. A convolutional autoencoder (3.6×10^5 parameters, 32 latent values for 1024 samples) trained on the real noise windows *only*; the reconstruction error is a morphology-independent anomaly score.

6.2 Regularisation, bias–variance and what the features say

Logistic regression with L_1 and L_2 penalties (Fig. 12) reaches a validation AUC of 0.927 with all features; the 5-fold cross-validated L_1 choice $C = 0.019$ keeps 18 of the 22 features and costs nothing in AUC, while for $C \lesssim 10^{-3}$ the L_1 model collapses much faster than L_2 (it sets coefficients to zero one by one). Trees (Fig. 12): the validation AUC of gradient boosting saturates at 0.938 for depth 4–6 while the training AUC keeps rising (0.936 → 0.956 from depth 1 to 12): beyond depth ~ 4 there is only variance. The learning curve (depth 4) starts at a training AUC of 1.0 against 0.925 on validation with 500 examples, and the two approach each other (0.952 / 0.938) at 3.2×10^4 examples without having converged: the feature models are data-limited, which is the practical argument for giving the network the raw series. The most important features by permutation importance are the 10 ms excess-power score and the energy in the 50–180 Hz bands; the first principal component of the features carries 46 % of the variance. Validation AUCs: ridge 0.927, random forest 0.938, gradient boosting 0.938.

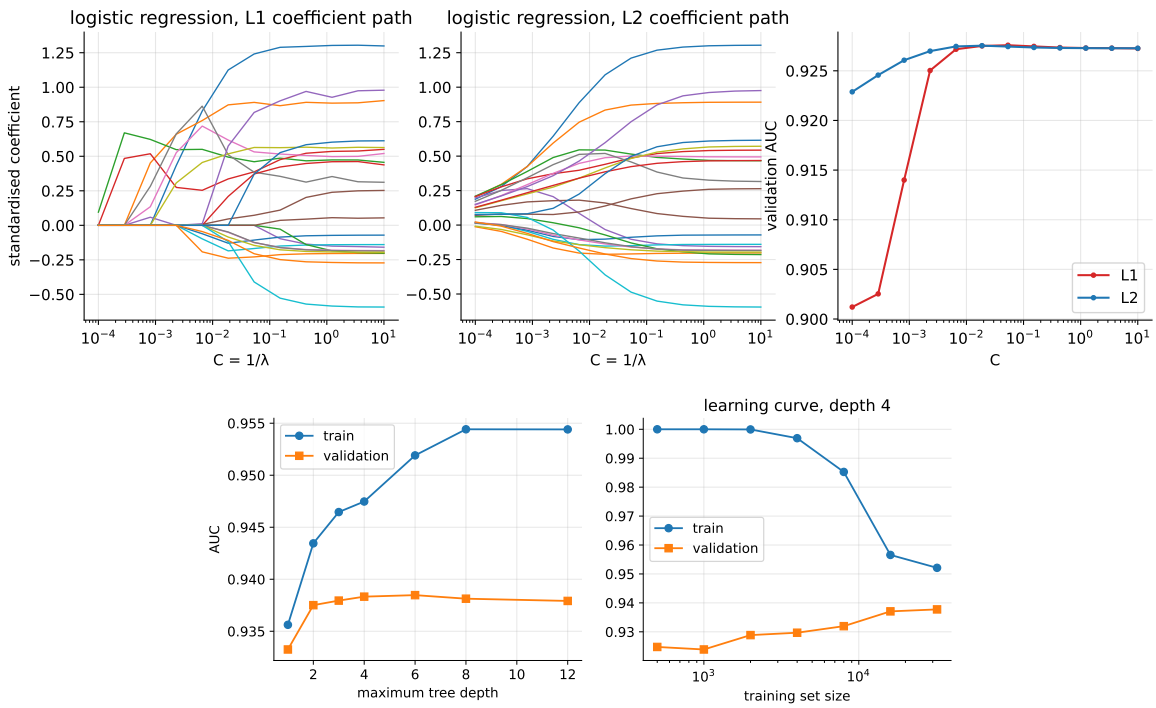


Figure 12: Top: L_1 and L_2 coefficient paths and validation AUC against C. Bottom: bias–variance (AUC against tree depth) and learning curve of gradient boosting.

6.3 Comparison with the frequentist statistics on the same injections

All statistics are evaluated on exactly the same test windows and injections, with the threshold of each set to the 99 % quantile of its own distribution on the *same* background, the validation noise (chunks 1 and 5). A common background is what makes the comparison fair: Section 4

used the training chunks, whose glitchy chunk 2 gives a more conservative bank threshold (there $\rho_{50} = 5.5$ for sine-Gaussians; here 4.9). Table 6 and Fig. 13 summarise the result; if the threshold is instead set on the test noise itself (equal observed FAP) the networks gain 0.4–0.6 in ρ_{50} (the test epochs are cleaner than the validation ones) and the bank only 0.04, so that the ensemble then also beats the bank in family (4.6 against 4.8); the conclusions for pulse and chirp and for the unseen morphology are unchanged.

	sine-Gaussian		Gaussian pulse		linear chirp	
statistic	ρ_{50}	ρ_{90}	ρ_{50}	ρ_{90}	ρ_{50}	ρ_{90}
matched-filter bank	**4.87**	**6.53**	5.34	6.82	6.28	8.21
excess power	6.54	9.10	5.77	7.54	10.10	12.21
ridge-logistic (22 features)	6.40	8.26	5.72	7.46	6.81	8.96
gradient boosting (22 features)	5.61	7.30	5.70	7.24	6.06	7.73
CNN (mixed training)	5.56	7.80	4.62	6.52	5.71	7.64
CNN ensemble (5)	5.05	7.05	**4.21**	**5.90**	**5.28**	**6.92**
CNN (sine-Gaussian only)	5.09	6.87	6.50	12.93	6.80	10.82
autoencoder, window MSE	12.09	15.14	12.44	15.06	12.32	15.14
autoencoder, local max	6.89	8.80	6.75	8.53	8.30	11.08

Table 6: Optimal SNR at 50 % and 90 % efficiency (FAP 10^{-2} on validation noise, test injections). Bold: best in each column.

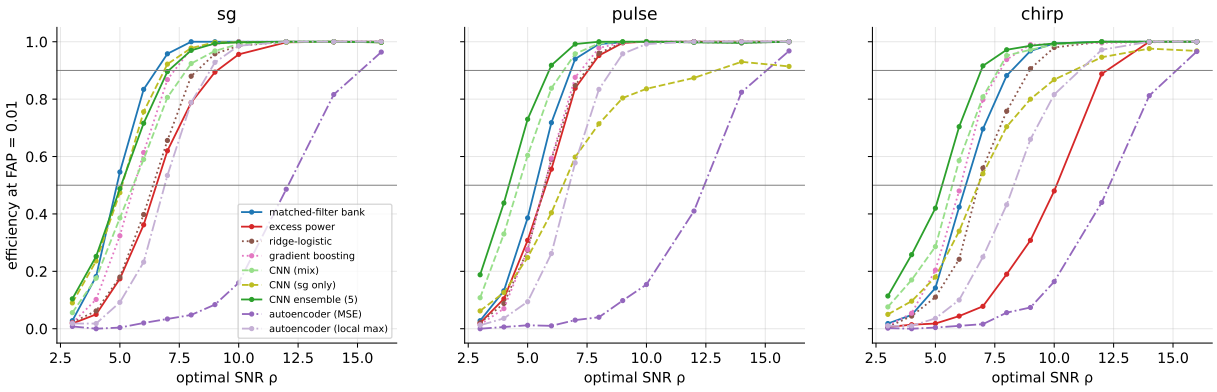


Figure 13: Efficiency against optimal SNR at FAP 10^{-2} for all statistics, on the same test injections.

- **In family (sine-Gaussian).** The matched-filter bank and the ensemble are close ($\rho_{50} = 4.9$ and 5.1); the bank is better at 90 % (6.5 against 7.1), the network wins by 0.2 when the threshold is set on the test noise. Parity is a respectable result for a network that has seen no template.
- **Morphologies in the training set (pulse, chirp).** The ensemble is about one unit of SNR better than the bank (4.2 against 5.3; 5.3 against 6.3): it has *learned* shapes that the bank covers only through its sine-Gaussian templates. Gradient boosting on 22 features is already at the bank level.
- **A morphology not seen in training.** The CNN trained on sine-Gaussians only loses efficiency exactly where its training stops: $\rho_{50} = 6.5$ (pulse) and 6.8 (chirp) against 5.3 and 6.3 for the bank, and the efficiency at $\rho = 8$ falls from 0.98 to 0.71 and 0.70 (bank: 0.99, 0.88). The advantage of a supervised detector is bought with training data; outside the training distribution the bank, which has no data to fail on, is the more robust.
- **Unsupervised.** The reconstruction error of the whole window is the total energy of a second of almost white noise — there is nothing to learn — and is inefficient ($\rho_{50} \approx 12$). The *local* maximum of the error recovers the performance of excess power (6.9, 6.8, 8.3) but not more: an autoencoder can beat an energy statistic only if the noise has structure that

it can learn, which after whitening it hardly has.

- **False alarms.** On the test chunks the observed FAP at a nominal 1 % is 0.85 % (bank), 0.92 % (excess power), 0.75 % (gradient boosting), 0.53 % (ensemble) and 0.35 % (single CNN): no statistic is immune to the epoch dependence of the background.

6.4 Calibration and uncertainty

A score is useful to a physicist if it is a probability. On balanced validation and test sets (noise plus mixed injections with SNR 3–15) all models are close to calibrated (Fig. 14): the expected calibration error is 0.7 % for gradient boosting, 2.2–2.4 % for the CNNs and 2.7 % for the ridge model on the test set. The CNNs are slightly *under*-confident at intermediate scores (observed signal fraction up to ≈ 0.15 above the prediction between 0.1 and 0.7). Temperature scaling [15] fitted on validation does not help ($T = 0.94$–1.02; ECE changes by at most 0.002): the miscalibration is not of the temperature type. For the CNNs the ECE is 0.8–1.2 % on the validation epochs they were tuned on and 2.2–2.4 % on the test epochs, so most of it is an *epoch shift*. The ensemble is better than a single network in every proper score (Brier 0.0350 against 0.0413, log-loss 0.110 against 0.128, AUC 0.9938 against 0.9909); the AUC on the test epochs is higher than on the validation ones for every model (ensemble 0.988 $\rightarrow$ 0.994), because the test chunks are cleaner: no sign of over-fitting to the training epochs.

The epistemic uncertainty is estimated with the ensemble [16] and with MC dropout [17] (30 stochastic passes). The ensemble spread falls with SNR (mean std 0.16 at $\rho \approx 4$, ≈ 0.001 for $\rho \geq 12$) and peaks at $p \approx 0.5$; MC dropout shows the same trend but is only moderately correlated with it (0.35): they measure different things (spread over dropout masks of one network against disagreement of independently trained networks). Rejecting the 20 % most uncertain windows raises the AUC from 0.9924 to 0.9997.

7 Comparison of the three approaches

Table 7 collects what each approach gives on the same data, and what it does *not* give.

What the comparison supports.

1. The frequentist and Bayesian statistics are two views of the same matched-filter cube and detect the same bursts; the Bayesian analysis adds calibrated posteriors and the prior dependence of the evidence, the frequentist analysis adds optimality statements (Neyman–Pearson oracle gap) and guaranteed coverage.
2. A network with no template knowledge reaches matched-filter performance on the family the bank was designed for, and exceeds it on the shapes it has seen. Its advantage is bounded by its training set: it is not a replacement for a morphology-independent statistic.
3. None of the statistics escapes the real noise: the glitch floor dominates the price of a low false-alarm probability, and the calibration of a threshold is a property of the epoch in which it was measured.

What it does not support. These are efficiencies for single-detector injections into one-second windows at a given *optimal SNR*; they are not astrophysical sensitivities, not rates, and say nothing about coincidence between detectors, where most of the real-world rejection of glitches comes from.

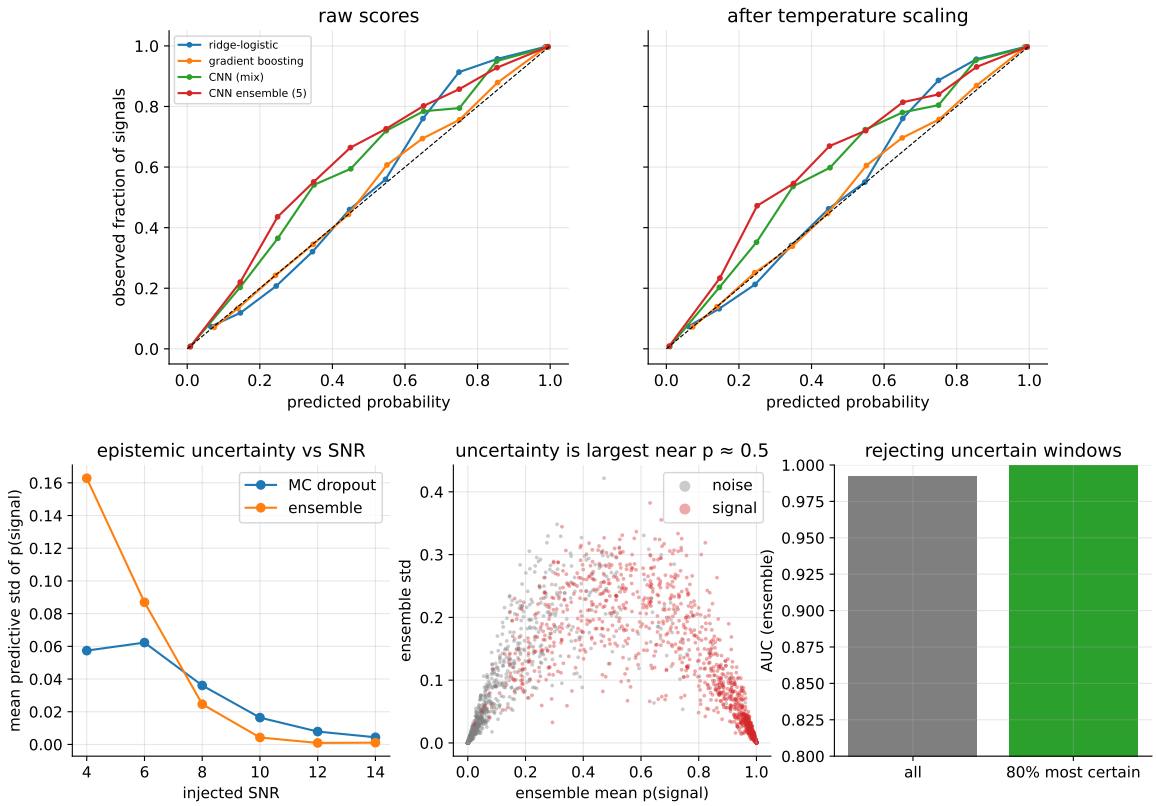


Figure 14: Top: reliability diagram on the test set, raw and after temperature scaling. Bottom: predictive standard deviation against SNR (ensemble and MC dropout), against the ensemble mean, and the effect of rejecting uncertain windows.

8 Conclusions and limits

Conclusions.

- **The real noise is the main result.** Whitened O1 H1 data are Gaussian in the bulk, but a glitch floor of a few 10^{-3} per second and strong epoch-to-epoch variations decide the price of a false-alarm probability: lowering it from 10^{-2} to 10^{-3} moves the 50 % point of the bank from $\rho \approx 5.5$ to ≈ 11 (in Gaussian noise the threshold would move only from ≈ 27 to ≈ 31, i.e. by ≈ 0.4 in SNR), and a threshold calibrated on one set of epochs is off by a factor ≈ 4.5 on others.
- **Frequentist.** The profile-likelihood bank reaches 50 % efficiency at $\rho \approx 5$ (oracle with known parameters: 3.0); the gap is the look-elsewhere penalty, quantified by $N_{\mathrm{eff}} \approx 6800$ independent trials. The MLE saturates the Fisher bound for $\rho \gtrsim 10$; the amplitude estimate is biased and the profile interval under-covers near the boundary, where Neyman/Feldman–Cousins intervals are exact. The dominant analysis systematic is the PSD estimator (mean against median Welch: $+0.025$ on $\varepsilon(\rho = 8)$).
- **Bayesian.** The exact grid posterior is validated against a closed form, MCMC and simulation-based calibration, in Gaussian and in real noise; the Bayes factor ranks windows like $\lambda_{\max}$ but is prior-dependent (Occam factor $-2 \ln \sigma_\rho$) and exceeds 100 on 0.2–0.9 % of pure-noise windows because of glitches. Credible intervals match confidence intervals for $\rho \gtrsim 3$ and differ, depending on the prior, below.
- **Machine learning.** A small CNN ensemble matches the bank on sine-Gaussians and beats it on the pulses and chirps it was trained on by about one SNR unit, but a network trained on one morphology fails on others; an autoencoder trained on noise adds nothing to an energy statistic. Calibration is good (ECE $\lesssim 3$ %), mostly limited by the epoch shift, and

	frequentist (bank)	Bayesian (same likelihood)	machine learning
statistic	$\lambda_{\max} = \max \rho^2$ (profile LR)	$\ln B_{10}$ (prior-weighted mean of the same cube)	CNN logit / ensemble mean
detection, sine-Gaussian	$\rho_{50} = 4.9$ (val. bkg)	identical to the bank within 0.01 in efficiency	5.1 (ensemble), parity
detection, other morphologies	5.3 / 6.3 (pulse / chirp), robust	as the bank	better if trained on them (4.2 / 5.3), worse if not (6.5 / 6.8)
needs	a template family	a template family and a prior	labelled injections; no template
parameters	MLE, saturates the Fisher bound for $\rho \gtrsim 10$	full posterior; calibrated (SBC)	none (not attempted)
uncertainty	profile LR / Neyman / FC intervals; coverage depends on SNR	credible intervals, prior-dependent at $\rho \lesssim 3$	ensemble / MC dropout; reject-uncertain gain
false alarms	calibrated on real background, epoch-dependent	$B_{10} > 100$ in 0.2–0.9 % of noise windows	calibrated on real background, epoch-dependent
cost	seconds	seconds (grid) – minutes (MCMC)	∼30 min training, ms inference

Table 7: Summary of the three approaches (FAP 10^{-2} per one-second window; SNR values are optimal SNRs).

ensemble uncertainty identifies the windows where the model should not be trusted.

Limits.

- **Single detector.** No coincidence and no time slides: the false-alarm numbers are per one-second window, not rates, and the glitch floor would be largely removed by a second detector.
- **Injections at a given SNR.** Not converted into distances or energies; the three morphologies are phenomenological and share the same t_0 prior.
- **Background statistics.** 1.6×10^4 training windows limit the lowest usable FAP to $\sim 10^{-3}$; the quoted 10^{-3} efficiencies carry a large uncertainty from the tail of the background.
- **Eight chunks.** The epoch dependence is measured with two test chunks; a systematic study of its size would need many more.
- **Grid resolution.** The Bayesian grid under-resolves f_0 and t_0 for $\rho \gtrsim 15$ (the SBC coverage of those two parameters, 0.50–0.60, reflects this, not the model).
- **ML numbers** are efficiencies *for the training injection prior*; the supervised detectors are only as good as the injections they have seen, and parameter estimation with ML was not attempted.
- **Comparison thresholds.** The bank is compared with the ML detectors on the validation background; the choice of background changes the numbers by up to 0.6 in ρ_{50} (Section 6).

Natural extensions. A second detector (L1) for coincidence and a real time-slide background; a rolling background close in time to the data; fine grids or nested sampling for the Bayesian part; training the networks with a glitch class (Gravity-Spy style) and with astrophysical waveforms; parameter regression with the CNN compared with the posterior of Section 5.

A Reproducibility and origin of the numbers

The project is a pipeline of notebooks in `Code_General/`, run in order with the Jupyter kernel `cern` (environment in `environment.yml`); every random number is generated with a fixed seed and long steps cache their output. `download_data.py` downloads the eight raw chunks (about 0.5 GB in float32; not tracked); `gwosc_tools.py` holds the shared functions (whitening, analytic waveforms, injection, template bank, statistics). Run times on a 10-core laptop: notebook 01 ≈1 min, 02 ≈1 min, 03 ≈10–15 min (most of it the systematic variants), 04 ≈20 min, 05 ≈30–40 min with training from scratch (a few minutes with the saved models).

report	notebook	result file
Sec. 2: data, PSD, Gaussianity tests	`01_data_whitening_validation`	`results/01_summary.json`
Sec. 3: waveforms, SNR, injection validation	`02_waveforms_and_injections`	`results/02_summary.json`
Sec. 4: bank, background, efficiency, MLE, toys, systematics	`03_frequentist_analysis`	`results/03_summary.json`, `results/03_systematics_raw.json`
Sec. 5: posterior, MCMC, Bayes factors, SBC, coverage	`04_bayesian_analysis`	`results/04_summary.json`
Sec. 6: features, CNN, autoencoder, calibration, uncertainty	`05_ml_detection`	`results/05_summary.json`
exploratory prototype (GW150914, Q-transform)	`00_waveform_exploration`	–

Table 8: Where each number of the report comes from.

The figures of the report are the `figures/fig0N_*.pdf` files written by the notebooks and copied into `Docs/report/figures/`. The report is compiled with `tectonic -X compile main.tex`.

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